Contents

1 Orbitals And Their Symmetry

We use the following order of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{rl}
b_{1u}\ \text{—} & \text{—}\ a_u \\
b_{2g}\ \text{—} & \text{—}\ b_{3g} \\
 & \\
b_{1u}\ \text{—} & \text{—}\ a_u \\
b_{2g}\ \text{—} & \text{—}\ b_{3g}
\end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:

$$\begin{array}{cc}
\text{—} & \text{—} \\
\text{—} & \text{—}
\end{array}$$

where the character of the orbital is defined by its position.
Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital order within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

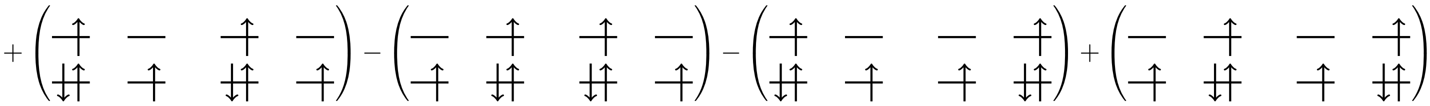
$$a_u^l = +b_{1g} + a_u \tag{1a}$$
$$b_{1u}^l = +a_g + b_{1u} \tag{1b}$$
$$b_{2g}^l = +b_{3u} + b_{2g} \tag{1c}$$
$$b_{3g}^l = +b_{2u} + b_{3g} \tag{1d}$$
$$a_u^r = -b_{1g} + a_u \tag{1e}$$
$$b_{1u}^r = -a_g + b_{1u} \tag{1f}$$
$$b_{2g}^r = -b_{3u} + b_{2g} \tag{1g}$$
$$b_{3g}^r = -b_{2u} + b_{3g} \tag{1h}$$

2 Monomer States and Configurations

$$
\begin{aligned}
i^3b_{2u} \quad &= +\begin{pmatrix} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} \qquad &= |a2a0| - |0a2a| \\
e^3b_{2u} \quad &= +\begin{pmatrix} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} \qquad &= |a2a0| + |0a2a| \\
e^3b_{3u} \quad &= +\begin{pmatrix} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} \qquad &= |aa20| - |02aa| \\
i^3b_{3u} \quad &= +\begin{pmatrix} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} \qquad &= |aa20| + |02aa|
\end{aligned}
$$

3 Linear Combinations: 16 Monomer- / 8 Dimer-States

identical i^3b_{2u} * i^3b_{2u} $\quad=\quad$ identical 1.3 * 1.3



written in CI Vectors:

$$+1\cdot|a2a0|\cdot|a2a0|-1\cdot|0a2a|\cdot|a2a0|-1\cdot|a2a0|\cdot|0a2a|+1\cdot|0a2a|\cdot|0a2a|$$

forms the following possible dimer combinations:

$$\begin{aligned}&+1\cdot|a2a0a2a0|+1\cdot|0a2a0a2a|-\frac{1}{8}\cdot|0a2aa2a0|+\frac{1}{8}\cdot|0a20a2aa|+\frac{1}{8}\cdot|0aaaa220|-\frac{1}{8}\cdot|0aa0a22a|\\&+\frac{1}{8}\cdot|022aaaa0|-\frac{1}{8}\cdot|0220aaaa|-\frac{1}{8}\cdot|02aaaa20|+\frac{1}{8}\cdot|02a0aa2a|+\frac{1}{8}\cdot|aa2a02a0|-\frac{1}{8}\cdot|aa2002aa|\\&-\frac{1}{8}\cdot|aaaa0220|+\frac{1}{8}\cdot|aaa0022a|-\frac{1}{8}\cdot|a22a0aa0|+\frac{1}{8}\cdot|a2200aaa|+\frac{1}{8}\cdot|a2aa0a20|-\frac{1}{8}\cdot|a2a00a2a|\end{aligned}$$

or, expressed in terms of symmetry:

$$\begin{aligned}=-4\left|\underbrace{a_gb_{2u}b_{1u}b_{3g}}_{a_g}\right|-2\left|\underbrace{a_gb_{3u}b_{2u}b_{1g}}_{a_g}\right|+2\left|\underbrace{a_gb_{3u}b_{3g}a_u}_{a_g}\right|+2\left|\underbrace{a_gb_{2u}b_{2g}a_u}_{a_g}\right|\\+2\left|\underbrace{a_gb_{1g}b_{2g}b_{3g}}_{a_g}\right|+2\left|\underbrace{b_{3u}b_{2u}b_{1u}a_u}_{a_g}\right|+2\left|\underbrace{b_{3u}b_{1g}b_{1u}b_{3g}}_{a_g}\right|+2\left|\underbrace{b_{2u}b_{1g}b_{1u}b_{2g}}_{a_g}\right|\\-2\left|\underbrace{b_{1u}b_{2g}b_{3g}a_u}_{a_g}\right|-4\left|\underbrace{b_{3u}b_{1g}b_{2g}a_u}_{a_g}\right|\end{aligned}$$

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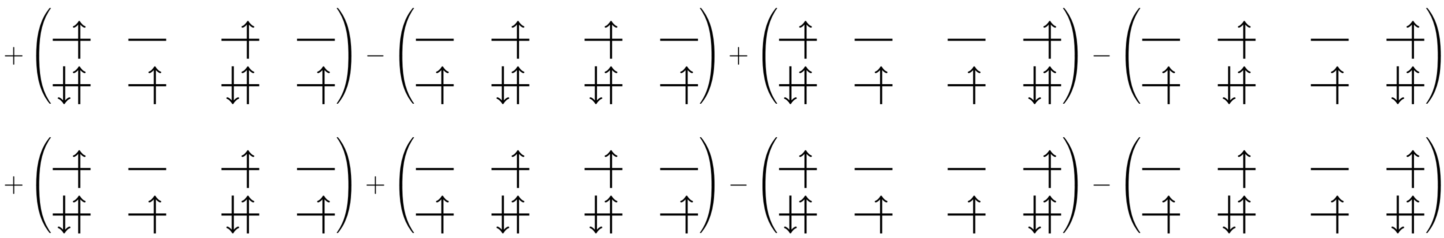
plus combination: i^3b_{2u} * e^3b_{2u} + e^3b_{2u} * i^3b_{2u} $\quad=\quad$ plus combination: 1.3 * 2.3 + 2.3 * 1.3



written in CI Vectors:

$$+1\cdot|a2a0|\cdot|a2a0|-1\cdot|0a2a|\cdot|a2a0|+1\cdot|a2a0|\cdot|0a2a|-1\cdot|0a2a|\cdot|0a2a|$$

$$+1\cdot|a2a0|\cdot|a2a0|+1\cdot|0a2a|\cdot|a2a0|-1\cdot|a2a0|\cdot|0a2a|-1\cdot|0a2a|\cdot|0a2a|$$

forms the following possible dimer combinations:

$$+2\cdot|a2a0a2a0|-2\cdot|0a2a0a2a|$$

or, expressed in terms of symmetry:

$$= \quad -8\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| \quad + \quad 8\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right|$$

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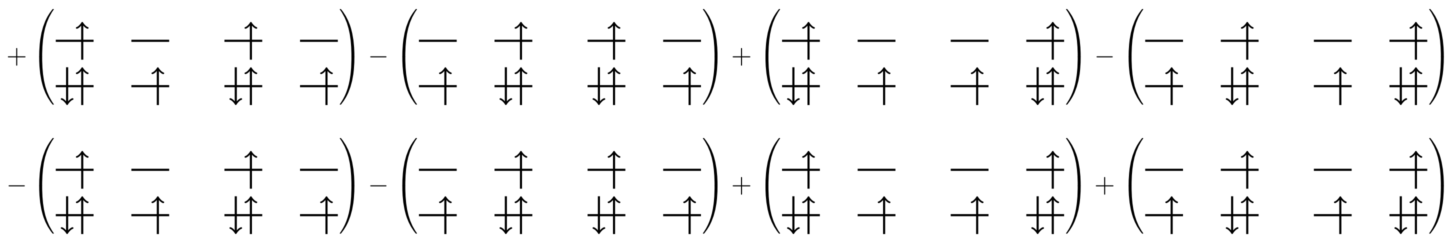
minus combination: $i^3 b_{2u}$ * $e^3 b_{2u}$ - $e^3 b_{2u}$ * $i^3 b_{2u}$ $\quad = \quad$ minus combination: 1.3 * 2.3 - 2.3 * 1.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a|$$

$$-1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

or, expressed in terms of symmetry:

$$= \quad -8\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| \quad + \quad 8\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right|$$

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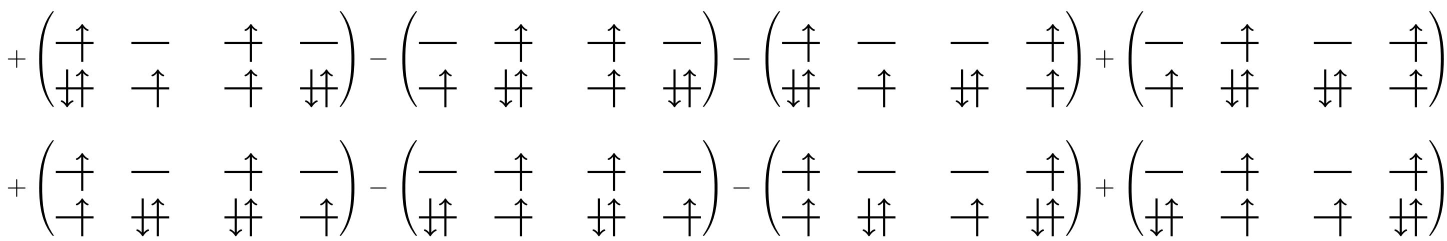
plus combination: $i^3 b_{2u}$ * $e^3 b_{3u}$ + $e^3 b_{3u}$ * $i^3 b_{2u}$ $\quad = \quad$ plus combination: 1.3 * 1.2 + 1.2 * 1.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa|$$

$$+1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| + \frac{1}{2} \cdot |0a2a02aa| + \frac{1}{2} \cdot |0aaa022a|$$
$$+\frac{1}{2} \cdot |022a0aaa| + \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4\left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - 4\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + 4\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$- 4\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + 4\left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - 4\left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + 4\left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

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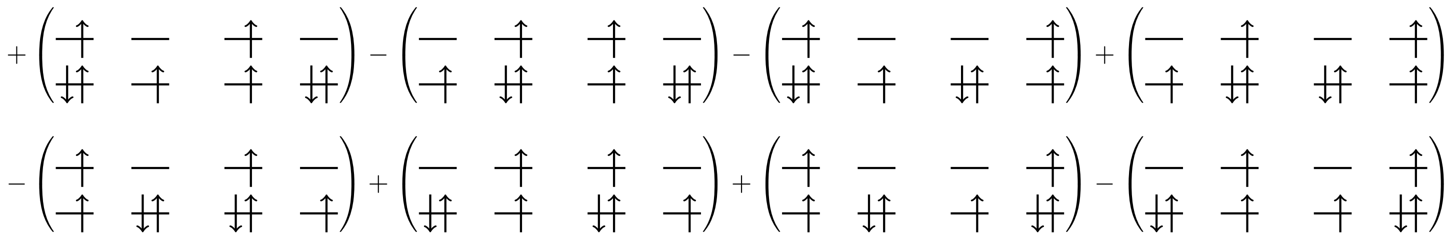
minus combination: i^3b_{2u} * e^3b_{3u} - e^3b_{3u} * i^3b_{2u} = minus combination: 1.3 * 1.2 - 1.2 * 1.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa|$$
$$-1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |0a2aaa20| - \frac{1}{2} \cdot |0a20aa2a| + \frac{1}{2} \cdot |aa2a0a20| - \frac{1}{2} \cdot |aa200a2a| + \frac{1}{2} \cdot |a2a002aa| - \frac{1}{2} \cdot |a2aa02a0|$$
$$+\frac{1}{2} \cdot |02a0a2aa| - \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$= -4\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4\left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - 4\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + 4\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$- 4\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + 4\left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - 4\left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + 4\left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

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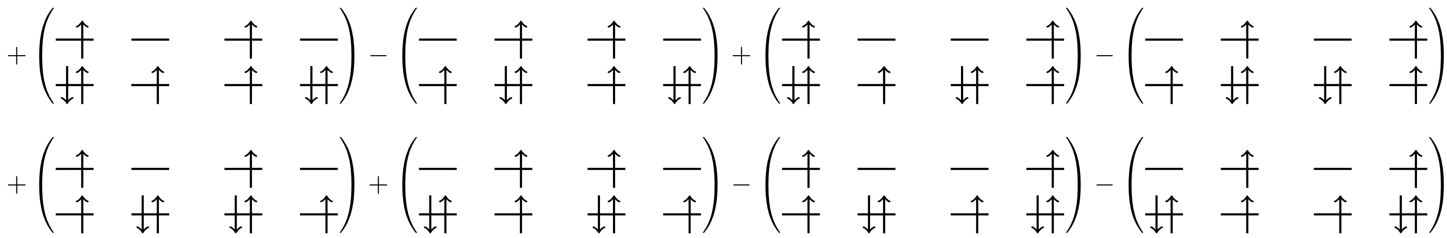
plus combination: $i^3b_{2u} * i^3b_{3u} + i^3b_{3u} * i^3b_{2u} \quad = \quad$ plus combination: 1.3 * 2.2 + 2.2 * 1.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$+1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

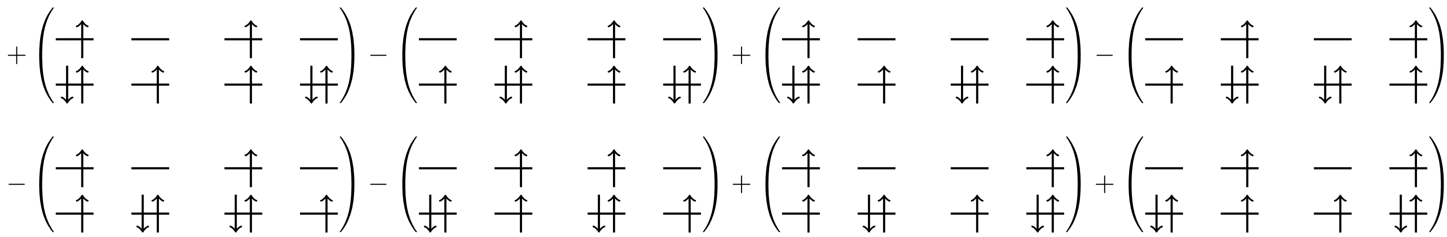
$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| - \frac{1}{2} \cdot |0a2a02aa| - \frac{1}{2} \cdot |0aaa022a|$$
$$-\frac{1}{2} \cdot |022a0aaa| - \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|$$
$$+ 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|$$

---

minus combination: $i^3b_{2u} * i^3b_{3u}$ - $i^3b_{3u} * i^3b_{2u} \quad = \quad$ minus combination: 1.3 * 2.2 - 2.2 * 1.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$-1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |0a2aaa20| - \frac{1}{2} \cdot |0a20aa2a| + \frac{1}{2} \cdot |aa2a0a20| - \frac{1}{2} \cdot |aa200a2a| - \frac{1}{2} \cdot |a2a002aa| + \frac{1}{2} \cdot |a2aa02a0|$$
$$-\frac{1}{2} \cdot |02a0a2aa| + \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$
\begin{aligned}
= -4 \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4 \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - 4 \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + 4 \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| \\
+ 4 \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - 4 \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + 4 \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - 4 \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|
\end{aligned}
$$

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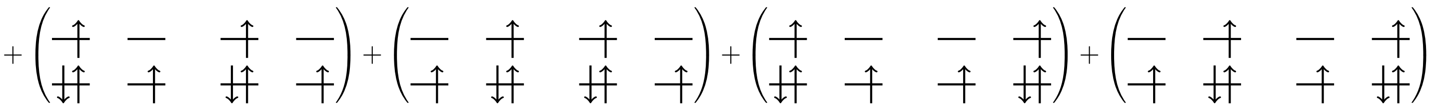
identical $e^3 b_{2u}$ * $e^3 b_{2u}$ = identical 2.3 * 2.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$
\begin{aligned}
&+1 \cdot |a2a0a2a0| + 1 \cdot |0a2a0a2a| + \frac{1}{8} \cdot |0a2aa2a0| - \frac{1}{8} \cdot |0a20a2aa| - \frac{1}{8} \cdot |0aaaa220| + \frac{1}{8} \cdot |0aa0a22a| \\
&-\frac{1}{8} \cdot |022aaaa0| + \frac{1}{8} \cdot |0220aaaa| + \frac{1}{8} \cdot |02aaaa20| - \frac{1}{8} \cdot |02a0aa2a| - \frac{1}{8} \cdot |aa2a02a0| + \frac{1}{8} \cdot |aa2002aa| \\
&+\frac{1}{8} \cdot |aaaa0220| - \frac{1}{8} \cdot |aaa0022a| + \frac{1}{8} \cdot |a22a0aa0| - \frac{1}{8} \cdot |a2200aaa| - \frac{1}{8} \cdot |a2aa0a20| + \frac{1}{8} \cdot |a2a00a2a|
\end{aligned}
$$

or, expressed in terms of symmetry:

$$
\begin{aligned}
= -4 \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + 2 \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - 2 \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - 2 \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| \\
- 2 \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - 2 \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - 2 \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - 2 \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| \\
+ 2 \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - 4 \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right|
\end{aligned}
$$

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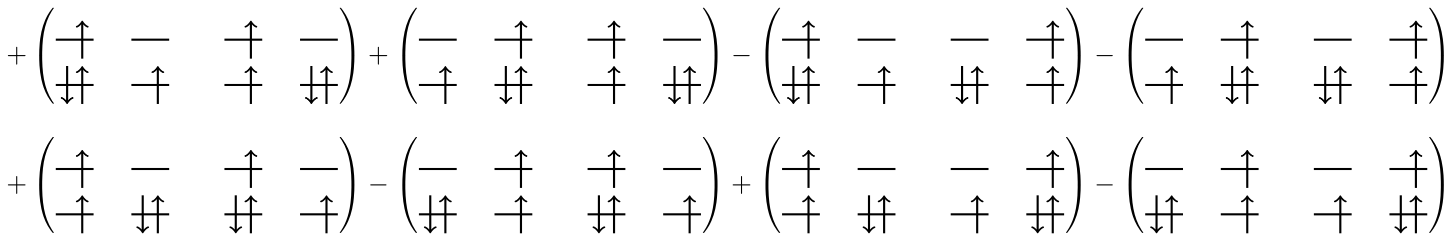
plus combination: e^3b_{2u} * e^3b_{3u} + e^3b_{3u} * e^3b_{2u} = plus combination: 2.3 * 1.2 + 1.2 * 2.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$+1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

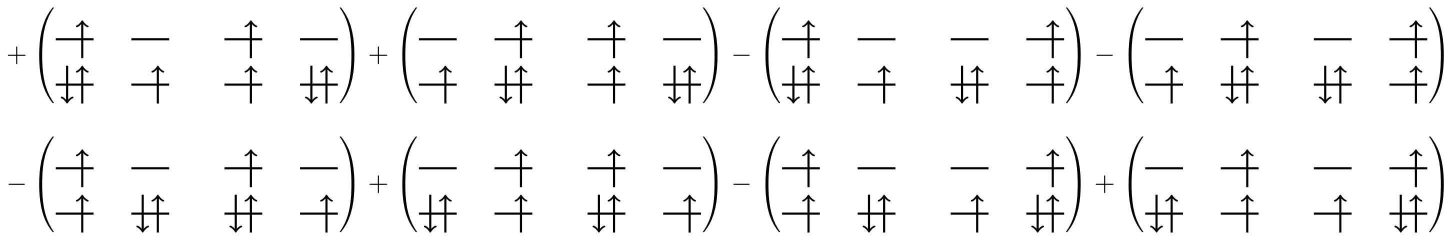
$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| - \frac{1}{2} \cdot |0a2a02aa| - \frac{1}{2} \cdot |0aaa022a|$$
$$-\frac{1}{2} \cdot |022a0aaa| - \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4 \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4 \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + 4 \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - 4 \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$- 4 \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + 4 \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + 4 \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - 4 \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

---

minus combination: e^3b_{2u} * e^3b_{3u} - e^3b_{3u} * e^3b_{2u} = minus combination: 2.3 * 1.2 - 1.2 * 2.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$-1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

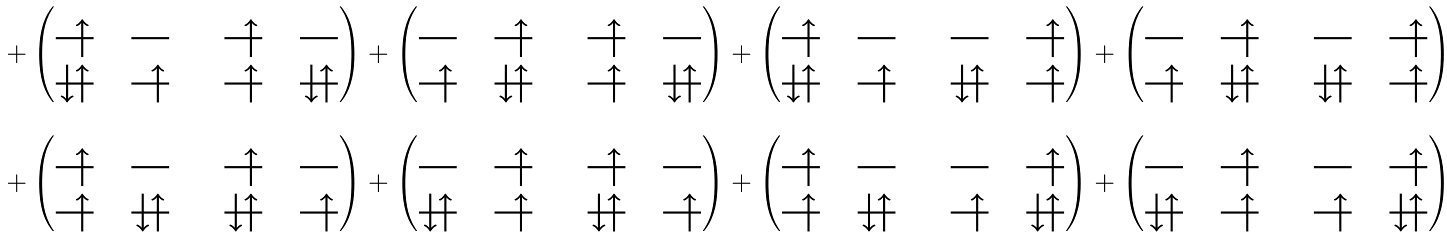
$$-\frac{1}{2} \cdot |0a2aaa20| + \frac{1}{2} \cdot |0a20aa2a| - \frac{1}{2} \cdot |aa2a0a20| + \frac{1}{2} \cdot |aa200a2a| + \frac{1}{2} \cdot |a2a002aa| - \frac{1}{2} \cdot |a2aa02a0|$$
$$+\frac{1}{2} \cdot |02a0a2aa| - \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$= -4\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4\left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + 4\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - 4\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$- 4\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + 4\left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + 4\left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - 4\left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

---

plus combination: $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$ = plus combination: 2.3 * 2.2 + 2.2 * 2.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa|$$
$$+1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| + \frac{1}{2} \cdot |0a2a02aa| + \frac{1}{2} \cdot |0aaa022a|$$
$$+\frac{1}{2} \cdot |022a0aaa| + \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4\left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + 4\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - 4\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$+ 4\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - 4\left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - 4\left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + 4\left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

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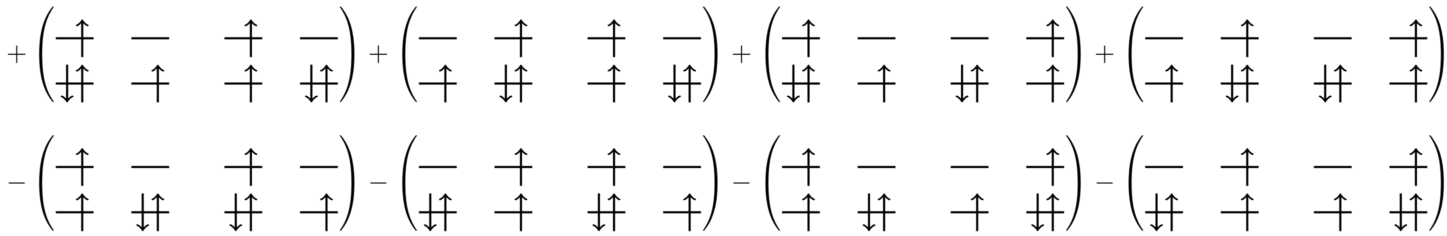
minus combination: e^3b_{2u} * i^3b_{3u} - i^3b_{3u} * e^3b_{2u} = minus combination: 2.3 * 2.2 - 2.2 * 2.3



written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa|$$

$$-1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$-\frac{1}{2} \cdot |0a2aaa20| + \frac{1}{2} \cdot |0a20aa2a| - \frac{1}{2} \cdot |aa2a0a20| + \frac{1}{2} \cdot |aa200a2a| - \frac{1}{2} \cdot |a2a002aa| + \frac{1}{2} \cdot |a2aa02a0|$$
$$-\frac{1}{2} \cdot |02a0a2aa| + \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$= -4 \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + 4 \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + 4 \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - 4 \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$
$$+ 4 \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - 4 \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - 4 \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + 4 \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

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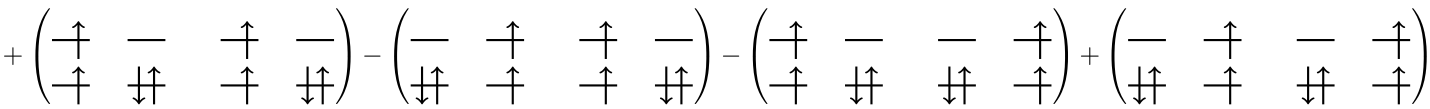
identical e^3b_{3u} * e^3b_{3u} = identical 1.2 * 1.2



written in CI Vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|$$

forms the following possible dimer combinations:

$$+1 \cdot |aa20aa20| + 1 \cdot |02aa02aa| - \frac{1}{8} \cdot |02aaaa20| + \frac{1}{8} \cdot |02a0aa2a| + \frac{1}{8} \cdot |022aaaa0| - \frac{1}{8} \cdot |0220aaaa|$$
$$+\frac{1}{8} \cdot |0aaaa220| - \frac{1}{8} \cdot |0aa0a22a| - \frac{1}{8} \cdot |0a2aa2a0| + \frac{1}{8} \cdot |0a20a2aa| + \frac{1}{8} \cdot |a2aa0a20| - \frac{1}{8} \cdot |a2a00a2a|$$
$$-\frac{1}{8} \cdot |a22a0aa0| + \frac{1}{8} \cdot |a2200aaa| - \frac{1}{8} \cdot |aaaa0220| + \frac{1}{8} \cdot |aaa0022a| + \frac{1}{8} \cdot |aa2a02a0| - \frac{1}{8} \cdot |aa2002aa|$$

or, expressed in terms of symmetry:

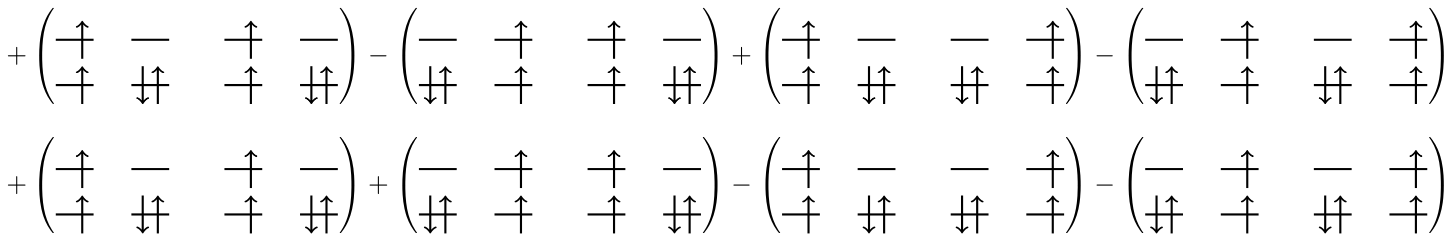
$$= -4\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - 2\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - 2\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - 2\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$+ 2\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + 2\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - 2\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - 2\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right|$$

$$- 2\left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - 4\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right|$$

---

plus combination: $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u}$ = plus combination: $1.2 * 2.2 + 2.2 * 1.2$



written in CI Vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|$$

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|$$

forms the following possible dimer combinations:

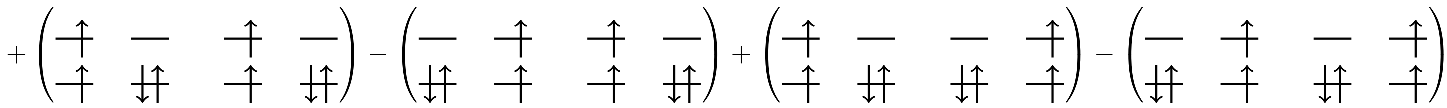
$$+2 \cdot |aa20aa20| - 2 \cdot |02aa02aa|$$

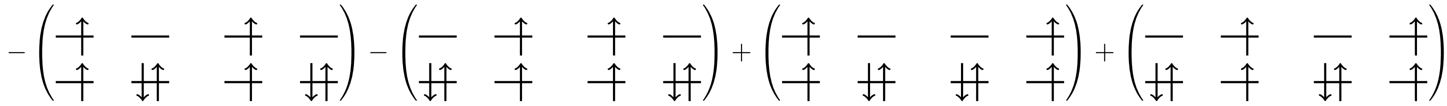
or, expressed in terms of symmetry:

$$= \quad -8\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| \quad + \quad 8\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right|$$

---

minus combination: e^3b_{3u} * i^3b_{3u} - i^3b_{3u} * e^3b_{3u} $\quad = \quad$ minus combination: 1.2 * 2.2 - 2.2 * 1.2





written in CI Vectors:

$$+1\cdot|aa20|\cdot|aa20| - 1\cdot|02aa|\cdot|aa20| + 1\cdot|aa20|\cdot|02aa| - 1\cdot|02aa|\cdot|02aa|$$

$$-1\cdot|aa20|\cdot|aa20| - 1\cdot|02aa|\cdot|aa20| + 1\cdot|aa20|\cdot|02aa| + 1\cdot|02aa|\cdot|02aa|$$

forms the following possible dimer combinations:

or, expressed in terms of symmetry:

$$= \qquad -8\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| \qquad + \qquad 8\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right|$$

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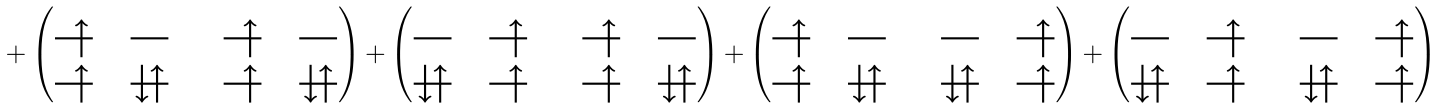
identical i^3b_{3u} * i^3b_{3u} $\quad = \quad$ identical 2.2 * 2.2



written in CI Vectors:

$$+1\cdot|aa20|\cdot|aa20| + 1\cdot|02aa|\cdot|aa20| + 1\cdot|aa20|\cdot|02aa| + 1\cdot|02aa|\cdot|02aa|$$

forms the following possible dimer combinations:

$$+1\cdot|aa20aa20| + 1\cdot|02aa02aa| + \frac{1}{8}\cdot|02aaaa20| - \frac{1}{8}\cdot|02a0aa2a| - \frac{1}{8}\cdot|022aaaa0| + \frac{1}{8}\cdot|0220aaaa|$$
$$-\frac{1}{8}\cdot|0aaaa220| + \frac{1}{8}\cdot|0aa0a22a| + \frac{1}{8}\cdot|0a2aa2a0| - \frac{1}{8}\cdot|0a20a2aa| - \frac{1}{8}\cdot|a2aa0a20| + \frac{1}{8}\cdot|a2a00a2a|$$
$$+\frac{1}{8}\cdot|a22a0aa0| - \frac{1}{8}\cdot|a2200aaa| + \frac{1}{8}\cdot|aaaa0220| - \frac{1}{8}\cdot|aaa0022a| - \frac{1}{8}\cdot|aa2a02a0| + \frac{1}{8}\cdot|aa2002aa|$$

or, expressed in terms of symmetry:

$$= -4\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + 2\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + 2\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + 2\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$
$$- 2\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - 2\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + 2\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + 2\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right|$$
$$+ 2\left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - 4\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right|$$